Set 19: Survey Responses Analysis

AIM

To analyze survey responses using R programming by creating a grouped bar chart, a stacked bar chart, and a table displaying survey responses.

ALGORITHM

1. Create a dataset containing Survey ID and responses to three questions.
2. Load the ggplot2 and dplyr libraries.
3. Convert the survey data into a suitable format for visualization.
4. Draw a grouped bar chart to visualize the distribution of answers for Question 1.
5. Create a stacked bar chart to represent the overall distribution of responses for all three questions.
6. Display the survey responses in a table.
7. Display the charts and table and interpret the results.

R PROGRAM

```
library(ggplot2)
library(dplyr)
library(tidyr)

survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

long_data <- pivot_longer(
  survey, cols = starts_with("Question"),
  names_to = "Question", values_to = "Response"
)

ggplot(filter(long_data, Question == "Question1"),
```

```
      aes(x = Response, fill = Response)) +
  geom_bar() +
  labs(title = "Distribution of Question 1 Responses",
       x = "Answer", y = "Count")


ggplot(long_data, aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(title = "Overall Survey Responses",
       x = "Question", y = "Count")


print(survey)
```

OUTPUT:



RESULT

The grouped bar chart displays the distribution of answers for Question 1. The stacked bar chart compares responses across all three questions, and the table displays the responses of all three survey participants.